

Introduction to Microeconomics

Elasticity

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x -elasticity of y

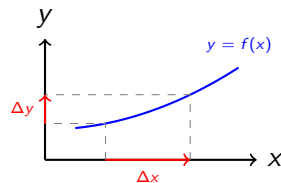
Consider a functional relationship between an endogenous variable y and an exogenous variable x , given by

$$y = f(x).$$

Then the x -elasticity of y is given by:

$$\epsilon = \frac{\Delta\%y}{\Delta\%x} = \frac{\frac{\Delta y}{y}}{\frac{\Delta x}{x}} = \left(\frac{\Delta y}{\Delta x} \right) \left(\frac{x}{y} \right).$$

- **Rate of Change:** The term $\frac{\Delta y}{\Delta x}$ represents the rate of change of f .
- **Percentage Change:** $\Delta\%y = \frac{y^1 - y^0}{y^0}$.



Price Elasticity of Demand

Price Elasticity of Demand

Percentage change in quantity demanded of a good resulting from a 1% increase in its **own price**.

$$\epsilon_p^d = \frac{\Delta\%q}{\Delta\%p} = \frac{\frac{\Delta q}{q}}{\frac{\Delta p}{p}} = \left(\frac{\Delta q}{\Delta p} \right) \left(\frac{p}{q} \right).$$

Example

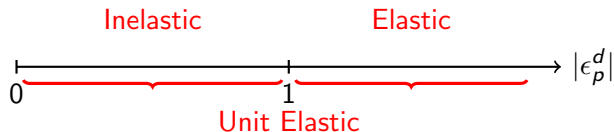
If a 1% rise in the price of wheat caused a 2% reduction in the quantity of wheat demanded, the price elasticity of demand for wheat would be -2 .

Attention!

The price elasticity of demand will always be non-positive because price changes always move in the opposite direction from changes in quantity demanded (the law of demand).

Dictionary of terminology and conventions

- Out of convenience, we usually drop the negative sign and speak of price elasticities in absolute values: $|\epsilon_p^d|$.
- The demand for a good is said to be:
 - **Elastic** with respect to its price if its price elasticity is more than 1.
 - **Inelastic** with respect to its price if its price elasticity is less than 1.
 - **Unitary Elastic** with respect to price if its price elasticity is equal to 1.



Price Elasticity of Demand

Recall the formula

$$\epsilon_p^d = \left(\frac{\Delta q}{\Delta p} \right) \left(\frac{p}{q} \right).$$

It has two components:

- ① $\frac{\Delta q}{\Delta p}$, needs to be computed from the demand function.
- ② $\frac{p}{q}$, needs a specific point (q, p) in the demand curve.

When in (2) above you pick a specific point (q, p) to compute the elasticity, we call it the **point-price elasticity of demand**.

A difficulty remains, how to compute (1)? You have two choices:

- You need a second point to compute an approximation of term (1)
- You need full information about the shape of the demand function $q^d(p)$ and use Calculus to compute the derivative with respect to price.

Price elasticity for linear demand curves

Suppose that $q^d(p) = \alpha - \beta p$; and suppose that $\Delta p = p_1 - p_0$. Then

$$\Delta q = q_1 - q_0 = (\alpha - \beta p_1) - (\alpha - \beta p_0) = -\beta(p_1 - p_0) = -\beta \Delta p.$$

Then

$$\frac{\Delta q}{\Delta p} = -\beta.$$

Recall that when graphing q^d , we graph the inverse demand curve

$$p = \frac{\alpha}{\beta} - \frac{1}{\beta} q^d.$$

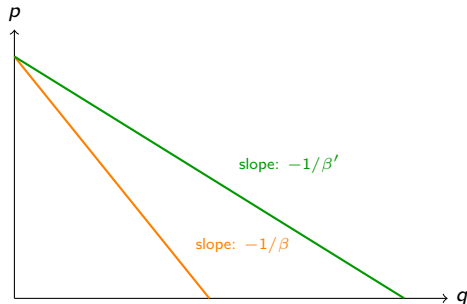
Thus, for a linear demand curve, the **point-price elasticity of demand at point (q, p)** is

$$\epsilon_p^d = -\beta \left(\frac{p}{q} \right) = -\frac{\beta p}{\alpha - \beta p}.$$

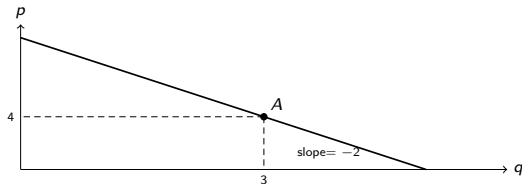
Notice that the elasticity is inversely proportional to the slope of the inverse demand curve.

Impact of Slope on Elasticity

Consider two demand curves starting at the same price intercept. If $\beta' > \beta$, then the flatter curve D' is more elastic at any given price.

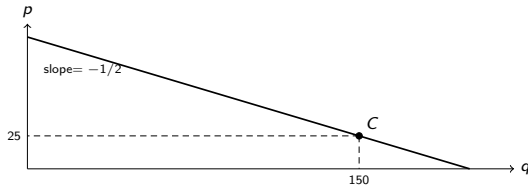


$$\beta' > \beta \implies |\epsilon_p^{d'}| > |\epsilon_p^d|$$



$$|\epsilon_p^D| = \left| \frac{\Delta q}{\Delta p} \cdot \frac{p}{q} \right| = \left| -\frac{1}{2} \cdot \frac{4}{3} \right| = \frac{2}{3}$$

at $(3, 4)$ the demand is inelastic.



$$|\epsilon_p^D| = \left| \frac{\Delta q}{\Delta p} \cdot \frac{p}{q} \right| = \left| -2 \cdot \frac{25}{150} \right| = 1/3$$

at $(150, 25)$ the demand is inelastic.

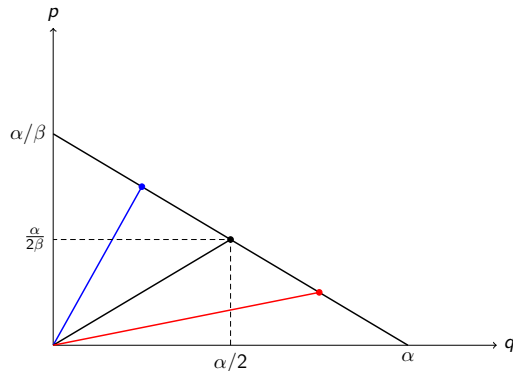
Discussion

Notice that in the expression for the point-price elasticity of a linear demand curve

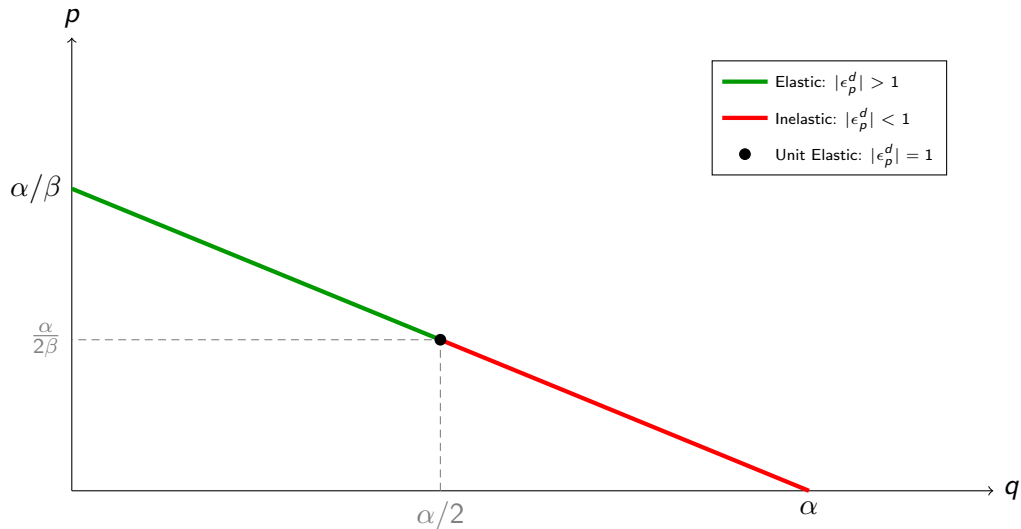
$$\epsilon_p^d = -\beta \frac{p}{q} = -\beta \frac{p}{\alpha - \beta p},$$

β is constant, but the ratio p/q is not; it actually declines as we move along the demand curve.

- The ratio p/q is the slope of a ray from the origin $(0,0)$ to a point (q, p) on the demand curve.
- As price falls and quantity increases, these rays become flatter.
- Therefore, $0 < |\epsilon_p^d| < \infty$ along the linear demand curve.
- Notice that at the midpoint $\left(\frac{\alpha}{2}, \frac{\alpha}{2\beta}\right)$, the slope of the ray is $1/\beta$, and thus, the elasticity is $|\epsilon_p^d| = 1$ at the midpoint.

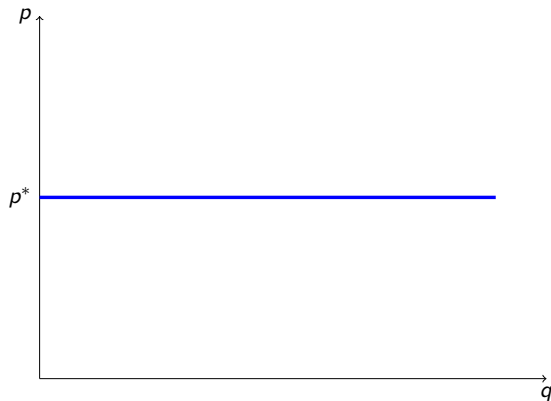


Elasticity Along a Linear Demand Curve



Infinitely Elastic Demand

- For a horizontal demand curve $\Delta q / \Delta p$ is infinite, thus $|\epsilon_p^d| \rightarrow \infty$.
- A tiny change in price triggers an enormous change in demand $\Rightarrow$ the elasticity of demand is infinite.
- Also called **Perfectly Elastic Demand**.
- Intuition: consumers will buy as much of a good as they can get at a single price, but for any higher price the quantity demanded drops to zero, while for any lower price the quantity demanded increases without limit.



Infinitely Inelastic Demand

- For a vertical demand curve $\Delta q / \Delta p$ is 0, thus $|\epsilon_p^d| = 0$.
- Because the quantity demanded is the same no matter what the price, the elasticity of demand is zero.
- Also called **Perfectly Inelastic Demand**.
- Intuition: Consumers will buy a fixed quantity of a good, regardless of its price.

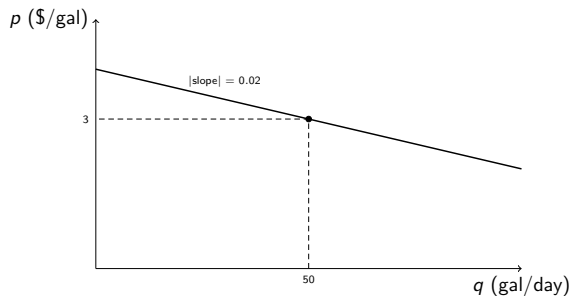


The unit-free property of Elasticity

Why measure price-responsiveness with elasticity, which seems complex, rather than with the slope, which seems simple?

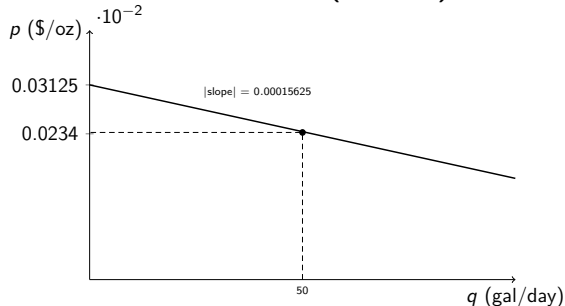
Elasticity is unit-free, whereas the slope is not.

Market for Milk (Gallons)



$$|\epsilon_p^d| = \left| \frac{1}{-0.02} \cdot \frac{3}{50} \right| = 3$$

Market for Milk (Ounces)



$$|\epsilon_p^d| = \left| \frac{1}{-0.00015625} \cdot \frac{0.0234375}{50} \right| = 3$$

Steepness $\neq$ Elasticity

- **Visual Trap:** We often assume flatter curves are more elastic.
- **Reality:** Elasticity depends on the *point* of evaluation.

Point A (Steep)

$$q = 1, \quad p = 8, \quad \beta = 0.5$$

$$\epsilon_p^d = (-\beta) \cdot \frac{p}{q}$$

$$= (-0.5) \cdot \frac{8}{1}$$

$$= -4$$

$$|\epsilon_p^d| = 4 \text{ (Elastic)}$$

Point B (Flat)

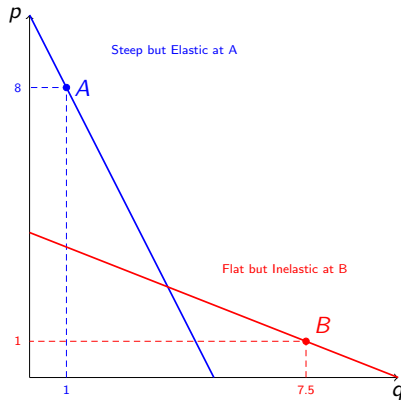
$$q = 7.5, \quad p = 1, \quad \beta = 2.5$$

$$\epsilon_p^d = (-\beta) \cdot \frac{p}{q}$$

$$= (-2.5) \cdot \frac{1}{7.5}$$

$$= -1/3$$

$$|\epsilon_p^d| = 1/3 \text{ (Inelastic)}$$



Case 1

Suppose that you are the administrator of the toll booths at a highway. At the current toll of \$1/trip, 100,000 trips per hour are taken across the highway. If the price elasticity of demand for trips is (in absolute value) 2, what will happen to the number of trips taken per hour if you raise the toll by 10%? What happens to total expenditures?

Quantity Calculation

- $\Delta\%p = 10\%$
- $|\epsilon_p^d| = 2$
- $\Delta\%q = -|\epsilon_p^d| \cdot (\Delta\%p) = -2 \cdot (10\%) = -20\%$
- New $q = 100,000 \cdot (1 - 0.20) = 80,000$ trips/hour

Revenue (Expenditure) Analysis

- Original Revenues
 $= 100,000 \cdot \$1.00 = \$100,000$
- New Revenues
 $= 80,000 \cdot \$1.10 = \$88,000$
- Since $|\epsilon_p^d| = 2 > 1$, demand is **elastic**.
- When demand is elastic:
 $\uparrow p \implies \downarrow \text{Revenues}$

Case 2

Now suppose that the price elasticity is 0.5. How would the number of trips and total revenues then be affected by a 10% increase in the toll?

Quantity Calculation

- $\Delta\%p = 10\%$
- $|\epsilon_p^d| = 0.5$
- $\Delta\%q = -0.5 \cdot (10\%) = -5\%$
- New $q = 95,000$ trips/hour

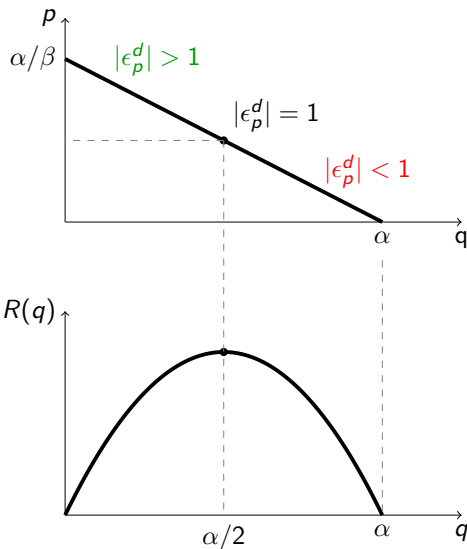
Revenue (Expenditure) Analysis

- New Revenues = $(95,000) \cdot (\$1.10)$
- New Revenues = \$104,500
- Since $|\epsilon_p^d| = 0.5 < 1$, demand is **inelastic**.
- When demand is inelastic:
 $\uparrow p \implies \uparrow \text{Revenues}$

Elasticity and Total Expenditure or Revenue

- A price reduction will increase total revenue if and only if the price elasticity of demand is greater than 1.
- An increase in price will increase total revenue if and only if the price elasticity of demand is less than 1.
- Revenues as a function of q are:
$$R(q) = p(q) \cdot q = \left(\frac{\alpha}{\beta} - \frac{1}{\beta} q \right) q.$$
- Revenues as a function of p are:
$$R(p) = p \cdot q(p) = p(\alpha - \beta p).$$

If demand is	$P \uparrow$	$P \downarrow$
Elastic $ \epsilon_p^d > 1$	$\downarrow R$	$\uparrow R$
Inelastic $ \epsilon_p^d < 1$	$\uparrow R$	$\downarrow R$



Mathematical Derivation of Revenue Maximization

Derivative of Total Revenue with respect to Price

$$R = p \cdot q(p)$$
$$\frac{dR}{dp} = p \cdot \frac{dq(p)}{dp} + (1) \cdot q(p) \quad [\text{Product Rule}]$$

Setting $\frac{dR}{dp} = 0$ for maximization:

$$p \cdot \frac{dq}{dp} + q = 0$$
$$\frac{p}{q} \frac{dq}{dp} + 1 = 0 \implies \epsilon_p^d + 1 = 0$$
$$|\epsilon_p^d| = 1$$

Conclusion: Total expenditure is maximized where demand is unit elastic.

- Raising price increases total revenue *only if* demand is sufficiently inelastic:

$$\uparrow P \Rightarrow \uparrow R \text{ only if } \left| \epsilon_p^d \right| < 1.$$

- Many pricing decisions, in particular by institutions, implicitly assume:

Quantity demanded changes little when price increases.

- That assumption is equivalent to assuming **inelastic demand**.
- If demand is elastic, then $\uparrow P \Rightarrow \downarrow R$ (revenue can fall).

Conclusion: Pricing decisions without an elasticity analysis are economically unjustified.

Determinants of Price Elasticity of Demand

What makes demand more or less elastic?

① **Substitution Possibilities:** The "Golden Rule" of elasticity.

- More substitutes $\implies$ **Higher Elasticity** ($|\epsilon_p^d| > 1$).
- *Market Definition:* "Coffee" is inelastic, but "Medium Cappuccino with Coconut Syrup From Fermento" is highly elastic.

② **Budget Share:** The "Sunk Cost" of small items.

- Larger share of income $\implies$ **Higher Elasticity**.
- You react to a 10% rise in rent, but likely ignore a 10% rise in the price of salt.

③ **Time Horizon:** The ability to pivot.

- Longer time $\implies$ **Higher Elasticity**.
- In the short run, you must pay for gas to get to work. In the long run, you buy a Tesla or move closer to the office.

Business Insight (using Economic Theory)

Firms use branding to reduce **perceived substitutability**, making the demand they face more **inelastic**, so that $\uparrow P \Rightarrow \uparrow R$.

Other demand elasticities with respect to its determinants

Income Elasticity of Demand

Percentage change in the quantity demanded resulting from a 1% increase in income.

$$\epsilon_I^d = \frac{\Delta\%q}{\Delta\%I} = \frac{\frac{\Delta q}{q}}{\frac{\Delta I}{I}} = \left(\frac{\Delta q}{\Delta I} \right) \left(\frac{I}{q} \right).$$

Cross-price elasticity of demand

Percentage change in the quantity demanded of one good resulting from a 1% increase in the price of another.

$$\epsilon_{p_i}^{d_j} = \frac{\Delta\%q_j}{\Delta\%p_i} = \frac{\frac{\Delta q_j}{q_j}}{\frac{\Delta p_i}{p_i}} = \left(\frac{\Delta q_j}{\Delta p_i} \right) \left(\frac{p_i}{q_j} \right).$$

Dictionary of terminology and conventions

- For the income elasticity (ϵ_I), we say the good is:
 - **Inferior** with respect to income if its income elasticity is negative.
 - **Independent** with respect to income if its income elasticity is zero.
 - **Normal** with respect to income if its income elasticity is positive.
 - Some authors/textbooks call a good **Superior** with respect to income if its income elasticity is greater than one.
- For the cross-price elasticity of good d_j with respect to the price of good q_i ($\epsilon_{p_i}^{d_j}$) we say they are:
 - **Complements** if the cross-price elasticity of j with respect to the price of i is negative.
 - **Substitutes** if the cross-price elasticity of j with respect to the price of i is positive.
 - **Neutral** if the cross-price elasticity of j with respect to the price of i is zero.

The Importance of the Sign in Elasticity

Unlike price elasticity of demand (where we often use absolute values), the **sign** of these elasticities tells the economic story.

Income Elasticity (ϵ_I^d)

- $\epsilon_I^d > 0 \implies$ **Normal Good**
 - $0 < \epsilon_I^d < 1$: Necessity
 - $\epsilon_I^d > 1$: Luxury / Superior
- $\epsilon_I^d < 0 \implies$ **Inferior Good**

Cross-Price Elasticity ($\epsilon_{p_i}^{d_j}$)

- $\epsilon_{p_i}^{d_j} > 0 \implies$ **Substitutes**
 - (e.g., $p_{\text{Coke}} \uparrow \implies q_{\text{Pepsi}} \uparrow$)
- $\epsilon_{p_i}^{d_j} < 0 \implies$ **Complements**
 - (e.g., $p_{\text{Coffee}} \uparrow \implies q_{\text{Sugar}} \downarrow$)

Theoretical Rule of Thumb

The *magnitude* tells you the strength of the relationship, but the *sign* tells you the nature of the relationship.

- We have studied how to compute the point elasticity.
- However, if:
 - We don't have access to the exact demand function OR,
 - We have inaccurate information, i.e., we only have a few points in the schedule OR,
 - Price changes under consideration are large.
- If we have **at least two points on the demand curve**, then we use the **arc-elasticity of demand**.

$$\bar{\epsilon}_p^d = \frac{\frac{\Delta q}{\bar{q}}}{\frac{\Delta p}{\bar{p}}} = \left(\frac{\Delta q}{\Delta p} \right) \left(\frac{\bar{p}}{\bar{q}} \right),$$

where the bar denotes the average of the two points:

$$\bar{p} = \frac{p_1 + p_2}{2} \quad \text{and} \quad \bar{q} = \frac{q_1 + q_2}{2}.$$

The "Direction" Problem: Why we use Arc Elasticity

Suppose demand is given by $q = 10 - p$ ($\beta = 1$). Consider a price change between points **A** and **B**:

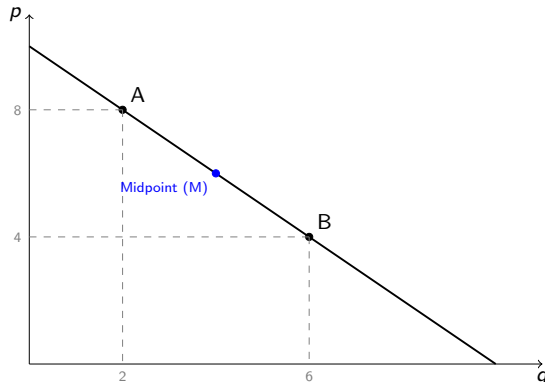
- **Point A:** $p_1 = 8, q_1 = 2$

$$|\epsilon_p^d|_A = 1 \cdot \frac{8}{2} = 4.0$$

- **Point B:** $p_2 = 4, q_2 = 6$

$$|\epsilon_p^d|_B = 1 \cdot \frac{4}{6} \approx 0.67$$

- **Problem:** If price drops from 8 to 4, which elasticity describes the change?



Solution: Arc Elasticity (Midpoint Method)

Using $\bar{p} = \frac{8+4}{2} = 6$ and $\bar{q} = \frac{2+6}{2} = 4$: $\bar{\epsilon}_p^d = \left(\frac{\Delta q}{\Delta p} \right) \left(\frac{\bar{p}}{\bar{q}} \right) = \left(\frac{6-2}{4-8} \right) \left(\frac{6}{4} \right) = (-1) \cdot (1.5) = -1.5 \Rightarrow |\bar{\epsilon}_p^d| = 1.5$.

Price Elasticity of Supply

Percentage change in quantity supplied (q) of a good resulting from a 1% increase in its **own price** (p).

$$\epsilon_p^s = \frac{\Delta\%q}{\Delta\%p} = \frac{\frac{\Delta q}{q}}{\frac{\Delta p}{p}} = \left(\frac{\Delta q}{\Delta p} \right) \left(\frac{p}{q} \right).$$

For a linear supply curve $q_s(p) = \gamma + \delta p$, the above expression simplifies to:

$$\epsilon_p^s = \delta \left(\frac{p}{q} \right).$$

The price elasticity of supply is always positive ($\epsilon_p^s > 0$), as price and quantity supplied move in the same direction (Law of Supply).

Notice that in the expression for the point-price elasticity of a linear supply curve

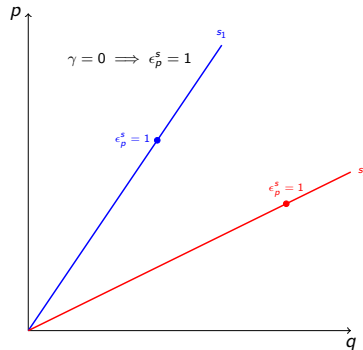
$$\epsilon_p^s = \delta \frac{p}{q},$$

d is constant, but what about the ratio p/q ?

- Consider a linear supply curve, which passes through the origin, i.e, $\gamma = 0$: $q_s = \delta p$.
- Substituting $q_s = \delta p$:

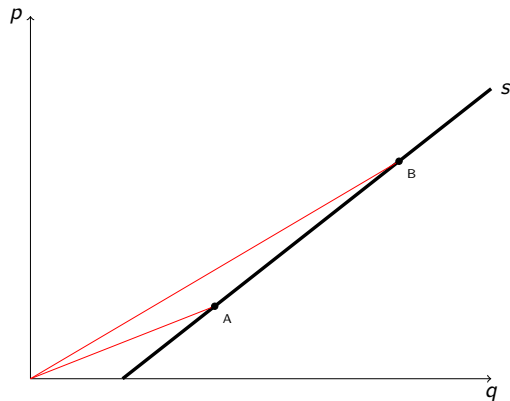
$$\epsilon_p^s = \delta \left(\frac{p}{\delta p} \right) = \frac{\delta p}{\delta p} = 1$$

- Any linear supply curve passing through the origin has a **unitary elasticity** ($\epsilon_p^s = 1$) at every point, regardless of δ .



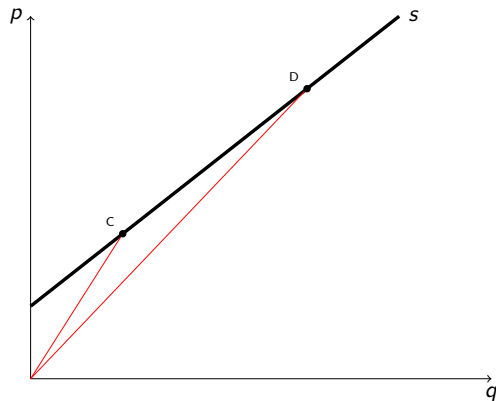
Geometry of Supply Elasticity

$$\gamma > 0$$



p/q is increasing along the supply curve, therefore ϵ_p^S is increasing along the curve.

$$\gamma < 0$$



p/q is decreasing along the supply curve, therefore ϵ_p^S is decreasing along the curve.

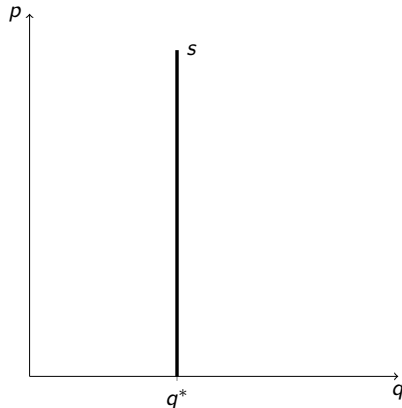
Infinitely Elastic Supply

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- A tiny change in price triggers an enormous change in supply $\Rightarrow$ the elasticity of supply is infinite.
- Also called **Perfectly Elastic Supply**.
- Intuition: producers will offer as much of a good as they can at a single price, but for any higher price, quantity supplied increases without limit, while for any lower price the quantity supplied drops to zero.



Infinitely Inelastic Supply

- For a vertical supply curve $\Delta q / \Delta p$ is 0, so $\epsilon_p^s = 0$.
- Because the quantity supplied is the same no matter what the price, the elasticity of supply is zero.
- Also called **Perfectly Inelastic supply**.
- Intuition: producers will offer a fixed quantity of a good, regardless of its price.



Determinants of Price Elasticity of Supply

What makes supply more or less elastic?

① Flexibility and Mobility of Inputs:

- Can inputs be easily shifted from one use to another?
- If a firm can readily reallocate workers or machines to different tasks, supply is more **elastic**.

② Availability of Substitute Inputs:

- If the production process requires a highly specialized, unique input (like a rare mineral or specialized talent), supply is **inelastic**.
- If the firm can use common, widely available inputs, supply is more **elastic**.

③ Time Horizon: The primary determinant.

- *Short Run*: At least one factor is fixed; supply is relatively **inelastic**.
- *Long Run*: All factors are variable; the firm can build new factories, making supply highly **elastic**.

Business Insight (using Economic Theory)

Firms with **excess capacity** or large **inventories** can respond to price spikes almost instantly (**elastic supply**). Conversely, "on-demand" manufacturing may leave a firm with **inelastic supply** in the short run, missing out on sudden revenue opportunities.